Staff Data Engineer

Abbott · United States > Madison : 1 Exact Lane

Job ID (this archive)	HI-042
Employer	Abbott
Location(s)	United States > Madison : 1 Exact Lane
Career stage (archive tag)	Experienced
Location group	Other U.S.
Majors (archive tags)	Computer Science, Software Engineering, Data Science
Pay listed	$99,300 – $198,700 per year
Employment type	Full time
Posted / listing date	2026-09-24
Employer requisition	31162159
Recruiting status when checked	Open when checked (Oct 7, 2026). Employer note: Readable; no closure notice found Saved PDF = full employer posting text, reformatted for print.
Source page	https://abbott.wd5.myworkdayjobs.com/abbottcareers/job/United-States--Madison--1-Exact-Lane/Staff-Data-Engineer_31162159-1
Captured	2026-10-07T20:10:54.311291 UTC
Course sheet refs	HI-042

> Captured on 2026-10-07 from the employer's public career site. The posting text below is the employer's own wording, reformatted for printing (site navigation removed). Live listings change and close; check the source page before applying.

Job description

Abbott is a global healthcare leader that helps people live more fully at all stages of life. Our portfolio of life-changing technologies spans the spectrum of healthcare, with leading businesses and products in diagnostics, medical devices, nutritionals and branded generic medicines. Our 122,000 colleagues serve people in more than 160 countries.

JOB DESCRIPTION:

The Staff Data Engineer is a senior, hands-on technical contributor within an Enterprise Data domain, owning technical design and the most complex implementation work for assigned data products, capabilities, and integrations. The role establishes, communicates, and evolves the technical approach for the work it leads and is accountable for the quality, durability, and supportability of the solutions it shapes.

Staff Data Engineers work directly with business stakeholders to understand needs and shape technical solutions, engaging at a level appropriate to the work they lead. The role is expected to be fluent in both technical execution and business context, translating between them without losing precision in either.

The Staff Data Engineer sets technical direction for assigned capabilities and initiatives within the domain, applies enterprise standards and platform patterns, engages Principal Engineers where cross-domain considerations apply, and multiplies the effectiveness of the domain team through design leadership, code review, and mentoring.

This role is based in Madison, WI. Relocation assistance may be available for qualified candidates.

Essential Duties

Include, but are not limited to, the following:

Technical design and solutioning

Own technical design and hands-on delivery for the most complex or highest-risk work across assigned data products, capabilities, and initiatives, including data models, pipeline architecture, integration patterns, and platform usage decisions.

Produce design documentation that allows others to understand, review, and build on technical decisions, and drive design review with domain and cross-domain peers.

Build and evolve assigned domain data products so they are documented, discoverable, governed, observable, and supported throughout their lifecycle for reuse across analytics, semantic layers, machine learning, and AI applications.

Define and uphold data contracts for the products the domain publishes, including schemas, freshness and availability expectations, breaking-change policy, and producer and consumer responsibilities.

Design data products for AI and machine learning consumption with reproducibility, lineage, timeliness, and defined data quality controls appropriate to the use case.

Diagnose and resolve complex production and data quality issues, and drive root cause resolution rather than recurring remediation.

Use approved AI-assisted development tooling where appropriate to accelerate engineering work, validating output against correctness, security, privacy, and quality standards.

Own the technical cost efficiency of assigned data products, including compute and warehouse sizing, job and query optimization, serverless and storage tradeoffs, and cost-to-serve implications.

Evaluate and recommend tools, patterns, and platform capabilities within enterprise standards, raising cases where an exception may be warranted.

Stakeholder engagement

Work directly with business stakeholders to understand needs, clarify requirements, and shape technical solutions.

Communicate technical concepts, tradeoffs, constraints, and delivery implications clearly to non-technical audiences.

Participate in technical working sessions with partner organizations such as Software Engineering, IT Applications, and Enterprise Architecture on integration and design questions affecting the domain.

Surface scope, priority, resourcing, and technical-debt implications of technical decisions for leadership review rather than resolving them independently.

Technical leadership and multiplication

Serve as a technical lead for assigned domain capabilities and initiatives, guiding engineers through design and implementation without formal authority.

Provide code and design review that raises quality and consistency across the domain team.

Mentor engineers and support their technical growth, including engineers transitioning into data engineering from adjacent disciplines.

Engage Principal Engineers on cross-domain architecture patterns and enterprise standards, and contribute domain perspective to initiatives that extend beyond the domain.

Lead adoption and continuous improvement of domain testing, observability, CI/CD, and delivery practices within enterprise standards.

Operational excellence and compliance

Ensure delivered solutions meet requirements for quality, reliability, scalability, performance, observability, security, privacy, access, and lifecycle management.

Design and implement appropriate handling of protected health information and other sensitive data, including access control, masking and de-identification, lineage, retention, and audit support consistent with HIPAA, CLIA, and enterprise privacy standards.

Serve as domain subject matter expert for significant production incidents affecting domain systems.

Uphold company mission and values through accountability, innovation, integrity, quality, and teamwork.

Support and comply with the company's Quality Management System policies and procedures.

Maintain regular and reliable attendance.

Ability to act with an inclusion mindset and model these behaviors for the organization.

Minimum Qualifications

Bachelor's degree in Data Science, Computer Science, Information Systems, Mathematics, or Engineering.

8+ years of progressively responsible experience in data engineering or a closely related discipline.

Demonstrated experience owning technical design and hands-on delivery for complex data products, pipelines, or platform capabilities from design through production support.

Advanced SQL and Python, with experience applying relational, dimensional, and semi-structured data modeling approaches. Experience designing and operating scalable, reliable distributed data systems on a modern data platform such as Databricks, including Spark, and using cloud infrastructure services such as AWS, Azure, or Google Cloud Platform.

Experience with version control, automated testing, and CI/CD practices, and with orchestration and transformation tooling such as Databricks Workflows, Delta Live Tables, Airflow, or dbt.

Experience delivering governed data using catalog, lineage, access control, and data quality capabilities such as Unity Catalog.

Demonstrated ability to work directly with business stakeholders to translate needs into technical solutions, mentor engineers, and raise technical quality across a team.

Demonstrated ability to perform the essential duties of the position with or without accommodation.

Preferred Qualifications

Experience with streaming and event-driven architectures such as Kafka.

Experience in a regulated environment such as HIPAA, CLIA, SOX, FDA, ISO 13485, or IEC 62304, ideally in life sciences, diagnostics, or clinical laboratory data handling protected health information.

Experience with semantic layers, metric definitions, data products, or feature pipelines supporting analytics, machine learning, and AI use cases in production.

Experience with open data file and table formats such as Parquet, Avro, and Delta Lake.

Experience with REST API development and integration patterns.

Familiarity with business intelligence concepts and semantic consumption patterns, including how data models and platform design affect performance in tools such as Tableau.

The base pay for this position is

$99,300.00 - $198,700.00 In specific locations, the pay range may vary from the range posted.

JOB FAMILY:

Product Development

DIVISION:

ONCO Cancer Diagnostics

LOCATION:

United States > Madison : 1 Exact Lane

ADDITIONAL LOCATIONS:

WORK SHIFT:

Standard

TRAVEL:

Yes, 5 % of the Time

MEDICAL SURVEILLANCE:

No

SIGNIFICANT WORK ACTIVITIES:

Continuous sitting for prolonged periods (more than 2 consecutive hours in an 8 hour day), Keyboard use (greater or equal to 50% of the workday)

Abbott is an Equal Opportunity Employer of Minorities/Women/Individuals with Disabilities/Protected Veterans.

EEO is the Law link - English: http://webstorage.abbott.com/common/External/EEO_English.pdf

EEO is the Law link - Espanol: http://webstorage.abbott.com/common/External/EEO_Spanish.pdf

Notes for ECS 1100 students

How this connects to your major

Data Science: healthcare data analysis, data engineering, statistics, AI or computational biology; Computer Science: healthcare software, embedded systems, automation or secure computing. These are teaching annotations, not an employer guarantee of eligibility.

Use the degree and experience evidence to distinguish a first job from a later career step.

DS: Named discipline appears in source; consult qualifications for eligibility

CS: Named discipline appears in source; consult qualifications for eligibility

Skills mentioned in the posting

Python, SQL, Git/source control, Machine learning, Data visualization, Cloud, Software testing, Medical-device quality

Course sheet tags

Majors (teaching tags): DS; CS; Career stage (teaching tag): Senior / leadership; Geography: Other US location; Recruiting status at capture: Posting readable at capture; no closure notice found; Employment type: Full time.

Source record

Original posting https://abbott.wd5.myworkdayjobs.com/abbottcareers/job/United-States--Madison--1-Exact-Lane/Staff-Data-Engineer_31162159-1

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